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UAV swarm safe coverage path planning with deep reinforcement learning

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Abstract

Unmanned aerial vehicles (UAVs) have been extensively deployed in various emergency rescue missions due to their cost-effectiveness, high maneuverability, and flexible flying capabilities. However, challenges such as densely packed towering buildings, no-fly zones, and complex urban environments significantly complicate rescue operations. To ensure equipment integrity and reliable data transmission, it is essential to implement an efficient coverage path planning strategy for the UAV swarm while prioritizing their safe landing. This paper addresses the problem of multi-UAV area coverage path planning in complex post-disaster urban scenarios, which is formulated as a maximum coverage rate optimization problem subject to safe landing rate requirements. Due to the uncertainties in the environment and UAV dynamics, a partially observable Markov decision process is adopted to model the flight status of UAVs. Accordingly, a deep reinforcement learning based collaborative strategy is developed, where collaborative rewards are determined through a coverage matrix. The approach prioritizes identifying the shortest coverage path while maximizing rewards, enabling real-time collaborative decision-making in UAV swarms. Comprehensive simulations are conducted, and the results show that the proposed method is effective for controlling UAV swarms. Furthermore, these results highlight the strong generalization capability of the proposed algorithm in dynamic environments.

Keywords Post-disaster emergency response, Reinforcement learning, Path planning, Unmanned aerial vehicles (UAVs) swarm

1 Introduction

Recently, unmanned aerial vehicles (UAVs) have increasingly emerged as one of the most promising solutions for various emergency rescue missions due to their low cost, high maneuverability, and unparalleled ability to reach areas that may be inaccessible or dangerous for human responders [1–3]. With the potential to reduce rescue losses and accelerate the rescue process, UAVs are becoming an essential tool for disaster prevention, mitigation, and emergency response [4–6]. However, in urban post-disaster rescue scenarios, there are still twofold technical challenges for UAV-based rescue. Firstly, the complex urban environment significantly complicates flight operations. High-rise buildings create narrow corridors with turbulent airflow patterns that can destabilize UAV

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attitude control. Additionally, collapsed structures form dynamic obstacles that severely constrain navigable airspace, while designated no-fly zones further limit feasible flight paths [7–10]. Traditional path planning algorithms often fail to adapt to rapidly changing conditions in real-time. Secondly, ensuring safe UAV operations is critical to preventing secondary disasters. This includes maintaining structural integrity, ensuring reliable data transmission, maximizing resource efficiency, and most importantly, avoiding crashes that could harm rescue personnel or survivors. Research indicates that post-disaster environments exhibit significantly higher landing failure rates due to debris interference and uneven landing surfaces. Uncontrolled crashes can block evacuation routes and cause casualties, thereby complicating rescue efforts [11, 12]. These factors collectively pose a significant challenge to the safe path planning of UAVs in emergency rescue operations. Unlike UAV-based emergency network coverage [13, 14], this paper highlights the scanning of affected areas to gather crucial information, such as videos or images, that depict the impacted environment.

Traditional UAV coverage path planning algorithms encompass both scanning and chunking methods. For example, the authors in [15, 16] introduced a versatile scanning method suitable for various region shapes, which includes interval one-way scanning, center scanning, and parallel scanning. Additionally, in [17], Xiao et al. developed a multi-UAV coverage path planning algorithm rooted in the chunking principle. Furthermore, as a decomposition-based coverage planning method, the scalable coverage path planning (SCoPP) algorithm employs cellular decomposition and auction mechanisms to handle non-convex regions with no-fly zones, enabling effective workload distribution among multi-UAV systems in post-disaster surveying scenarios [18]. However, these traditional algorithms predominantly rely on pre-known environments, posing significant challenges when adapting to complex and dynamically unknown scenarios, such as those encountered in search and rescue operations. Unlike SCoPP [15], which relies on prior static maps and auction-based task allocation to balance workload and ensure coverage completeness, and GLRL methods [19], which implicitly encourage safe return via energy consumption penalties, this work enables online decision-making under partial observability and explicitly models safe landing as a hard constraint that determines mission validity. Moreover, we emphasize a safety-coupled evaluation by reporting not only coverage rate but also safe landing rate and landing-conditioned coverage (CRAL), so that coverage obtained without safe landing is not credited. This distinction is central to our post-disaster setting, where mission success must simultaneously satisfy both effective coverage and safe landing.

Given the emerging challenges in coverage path planning applications, including environmental dynamics, high-dimensional problems, uncertainty, and real-time constraints, researchers have proposed deep learning-and reinforcement learning-based algorithms to enhance adaptability. For example, a UAV coverage path planning control method based on double deep Q-learning was proposed in [20]. A coverage path planning method based on map information was presented to optimize communication paths and improve communication network performance and coverage [21]. A multi-agent deep deterministic policy gradient (MADDPG)-based algorithm is proposed in [22] to jointly optimize the hovering and flight energy consumption of UAVs in an integrated manner, thereby significantly enhancing overall energy efficiency. Furthermore, [23] develops a dual-level optimization framework for autonomous UAV swarm coordination, which

effectively addresses mission planning challenges in disaster scenarios where global communication links are unavailable by enabling robust decentralized decision-making. This dynamic mission planning approach employs a hierarchical decision architecture to maintain operational continuity under disrupted network conditions. In parallel, deep reinforcement learning (DRL) has been increasingly adopted to tackle UAV-aided network optimization problems under complex constraints. For instance, priority-aware path planning and user scheduling in UAV-mounted MEC networks are investigated by jointly optimizing the UAV access path and offloading strategy under UAV battery capacity and flight-duration constraints [24]. Similarly, priority-aware resource scheduling for UAV-mounted MEC considers time-varying task priorities and optimizes hotspot selection and task offloading via deep Q-learning [25]. Beyond MEC scheduling, battery-constrained federated edge learning (FEEL) in UAV-enabled IoT networks is studied by jointly allocating CPU frequency and wireless bandwidth to balance latency and energy consumption, aiming to prolong battery life and avoid premature dropout [26]. Moreover, integrated communication and sensing in space-air-ground networks has been formulated as a joint optimization of UAV trajectory, time allocation, and data upload scheduling under energy and memory constraints [27]. Despite these advances, the above studies mainly emphasize communication/computation utility or data collection efficiency, and do not explicitly address the post-disaster rescue setting where the mission objective is rapid coverage of multiple discrete, irregularly distributed disaster regions/points while guaranteeing safe landing after task completion. Therefore, there remains a critical gap in jointly optimizing multi-UAV cooperative coverage efficiency over discrete disaster-affected clusters and landing safety constraints in dynamic urban post-disaster environments. Motivated by this gap, this work formulates a multi-UAV safe coverage path planning problem with an explicit safe-landing (rate) constraint and proposes a collaborative DRL framework to achieve efficient coverage while ensuring landing safety.

This paper addresses the issue of multi-UAV area coverage and safe path planning in complex urban post-disaster environments. The problem is formulated to achieve maximum coverage while adhering to safe landing rate constraints. Due to the uncertainties in the environment and UAV dynamics, a partially observable Markov decision process (POMDP) is adopted to model the flight status of UAVs. Although the static environment map D is assumed known a priori (e.g., GIS), the affected-area layer $\Phi(t)$ and coverage layer $W(t)$ are initially unknown and are progressively revealed through onboard sensing. Hence, the decision process is modeled as a POMDP with belief/partial observations. This study proposes a novel deep reinforcement learning approach grounded in collaborative strategies. In this framework, collaborative rewards are computed using the coverage matrix, prioritizing the shortest coverage path and assigning the highest reward to facilitate real-time collaborative decision-making among UAV swarms. Simulation experiments are conducted to validate the framework, with comparisons against previous methods. The effects of key parameters on UAV performance are systematically evaluated. Results indicate that the proposed method ensures efficient and safe search task completion in disaster areas through swarm control.

The main contributions of this paper are summarized as follows:

(1) We formulate a multi-UAV safe coverage path planning problem for post-disaster urban environments and adopt a safety-aware evaluation metric that couples coverage with safe landing.
(2) We propose a DDQN-based collaborative strategy reinforcement learning framework that integrates multi-channel map features with coverage-aware collaborative rewards to encourage efficient division of labor among UAVs.
(3) We conduct extensive simulations on City32 and City50 scenarios and compare with representative baselines to demonstrate improvements in landing success and coverage effectiveness under complex obstacle/no-fly constraints.

The remainder of this paper is organized as follows. Section 2 presents the environment and UAV models and formulates the safe coverage problem. Section 3 introduces the proposed reinforcement learning framework and collaborative reward design. Section 4 describes the neural network architecture and training details. Section 5 reports simulation results and comparisons. Section 6 concludes the paper and discusses future work.

2 Problem formulation

2.1 Environment model

In this paper, the urban environment is conceptualized as a three-dimensional (3D) cubical grid model of dimensions $I \times J \times H$, including designated start/landing positions for UAVs, restricted no-fly zones, and damaged buildings as obstacles. The 3D space is segmented into $I \times J \times H$ cells, with each horizontal cell denoted by Δij $(i \in \{1, 2, \ldots, I\}, j \in \{1, 2, \ldots, J\})$.

- Start/Landing positions $B = \{b_{11}, b_{12}, \ldots, b_{ij}\}$, where the variable $b_{ij} = 1$ denotes cell Δij as the designated start/landing position.
- No-fly zones $\Omega = \{\omega_{11}, \omega_{12}, \ldots, \omega_{ij}\}$, where the variable $\omega_{ij} = 1$ denotes cell Δij as a no-fly zones, otherwise 0.
- Obstacles. We define an obstacle height map $H^{obs} = \{h_{ij}\}$, where $h_{ij} \geq 0$ denotes the obstacle height (e.g., building height) at cell Δ_{ij}. The obstacle indicator is $\Psi = \{\psi_{ij}\}$ with $\psi_{ij} = 1$ if $h_{ij} > 0$, otherwise $\psi_{ij} = 0$. The obstacle footprint (length/width) is represented by the set of occupied cells $\{\Delta_{ij} : \psi_{ij}\}$.

Although the environment is represented as a 3D grid, $I \times J \times H$ the motion decision is formulated as 2D grid translation on the x–y plane with discrete altitude adjustment actions (Ascend/Descend).

2.2 UAV model

A swarm of UAVs is represented by the set $U = (U_1, U_2, \ldots, U_n)$, where U_g denotes the *gth* UAV, where $g \in \{1, 2, \ldots, n\}$. The UAV operates at an altitude $\hbar (0 \leq \hbar \leq H)$, with H representing its maximum flight height. To optimize the UAV's coverage detection capability, it operates at a lower altitude $\hbar$. When encountering an obstacle at cell Δ_{ij} with height h_{ij}, the UAV has three possible movement strategies, as follows

- If $h_{ij} < \hbar$, fly directly over the obstacle.
- When $H > h_{ij} \geq \hbar$, the UAV adjusts its altitude to navigate over the obstacle.
- If $h_{ij} \geq H$, avoid the obstacle.

For clarity, we define the feasible landing set as

$$\mathcal{B} \triangleq \{(i,j) \mid b_{ij} = 1,\ \omega_{ij} = 0,\ \psi_{ij} = 0\},$$

where b_{ij} indicates designated landing zones, and ω_{ij} and ψ_{ij} denote no-fly and obstacle cells, respectively. A trajectory is counted as safely landed if and only if the terminal position satisfies $(i_T, j_T) \in \mathcal{B}$ and the remaining battery is non-negative at termination. Accordingly, the safe-landing indicator can be equivalently written as

$$l_k = \mathbb{I}\{(i_T, j_T) \in \mathcal{B}\} \cdot \mathbb{I}\{B^l \geq 0\}.$$

This explicitly shows that safe landing is modeled as a feasibility requirement at the mission level, rather than a purely reward-based preference.

The UAV starts from the initial position $p_0 = (u_x^0, u_y^0, 0)$ and the initial battery is denoted by B_0. At time slor *t*, its real-time position of the UAV is denoted by $p_t = (u_x^t, u_y^t, \hbar^t)$ and the the remaining battery is denoted by B^t. Upon completing the disaster area scanning task, the UAV returns to its initial position with remaining battery denoted as B^l. The mission consists of *N* sequential tasks, indexed by $k \in \{1, \ldots, N\}$, and a binary variable l_m is introduced to indicate the landing safety after *m*th task. In the disaster area scanning task, $L = \{l_m\}$ records the landing outcome of each mission, where $l_m \in \{0, 1\}$ indicates whether the k-th mission ends with a safe landing. When the k-th task is completed, the landing is regarded as safe only if the UAV lands with sufficient remaining battery $B_m^l \geq 0$, and lands at a feasible landing cell, i.e., $b_{ij} = 1$, $w_{ij} = 0$,and $\psi_{ij} = 0$ (designated start/landing cell, not in a no-fly zone, and not an obstacle cell). In this case, $l_k = 1$; otherwise it is 0. The coverage area during the flight is represented by $C = \{C_{11}, C_{12}, \ldots, C_{ij}\}$, where cell Δij is covered by $C_{ij} = 1$, and not covered by $C_{ij} = 0$. The affected target area is the set $\Phi = \{\phi_{11}, \phi_{12}, \ldots, \phi_{ij}\}$where the variable $\phi_{ij} = 1$, i.e. cell Δij is the affected region.

3 Problems of safe path planning in urban environments

In this work, "safe landing" is modeled at the mission-planning level rather than at the low-level flight control layer. Specifically, safety is defined as the capability of a UAV to complete its assigned coverage task and return to a designated landing zone before battery depletion, while avoiding no-fly zones and impassable obstacles. Accordingly, this paper investigates safe path planning for multi-UAV swarms operating in urban post-disaster environments, with emphasis on autonomous takeoff, obstacle-aware navigation, effective area coverage, and safe return to the initial deployment locations, as illustrated in Fig. 1. The overall mission objective is to maximize the coverage of the disaster-affected area under limited flight time, collision avoidance constraints, and energy availability, while ensuring that all UAVs can safely return to their takeoff and landing zones. To capture the energy cost of UAV mobility, the energy consumption for traversing one grid cell is modeled as *b* for horizontal movement and ξb for vertical movement, where ξ is a model-dependent coefficient reflecting the additional energy required for altitude changes. Consequently, the energy consumed by a UAV when moving from time slot $t-1$ to time slot *t* can be expressed as follows:

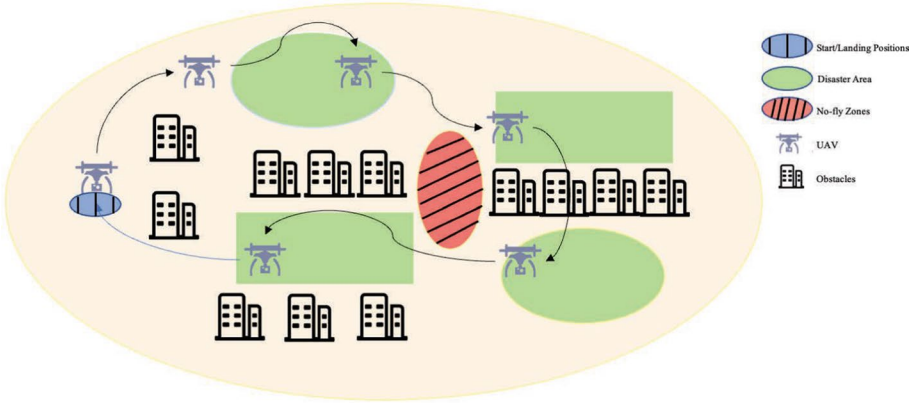


Fig. 1 UAV swarm path planning in urban post-disaster scenarios

$$b_t = b\left[\sqrt{\left(u_x^t - u_x^{t-1}\right)^2 + \left(u_y^t - u_y^{t-1}\right)^2} + \xi\left(\hbar^t - \hbar^{t-1}\right)\right] \tag{1}$$

It is worth noting that the energy consumption model in (1) adopts a high-level abstraction that captures the dominant effects of horizontal displacement and altitude variation on the UAV battery depletion. Although complex urban environments may induce turbulent airflow patterns and additional aerodynamic drag, explicitly modeling such effects would require detailed vehicle-specific parameters and significantly increase the state dimensionality, which is beyond the scope of this work.

Using the mobile energy consumption and the initial UAV battery B_0 from Eq. (1), the remaining battery when the UAV lands can be calculated as:

$$\mathcal{B}^l = \mathcal{B}^0 - \sum_{t=1}^{T} b_t \tag{2}$$

where B^l represents the battery power of the UAV at the time of landing , and T is the total execution time steps, with t ranging from 1 to T. Given UAV battery constraints, achieving full area coverage is challenging, necessitating a safe return and landing after maximum possible coverage. For N coverage missions, the average landing ratio β is expressed as follows:

$$\beta = \frac{1}{N} \times \sum_{k=1}^{N} l_k \tag{3}$$

here l_k is the safe-landing indicator defined in Sect. 2.2, which jointly considers remaining battery and landing-point feasibility (landing cell / no-fly / obstacle constraints). The coverage ratio characterizes the proportion of the disaster-affected area that is successfully covered by the UAV. Specifically, the collection ratio (CR) for the *kth* coverage mission, denoted by α_k is expressed as follows:

$$\alpha_k = \frac{\sum_{i=1}^{I}\sum_{j=1}^{J} C_{ij}\phi_{ij}}{\sum_{i=1}^{I}\sum_{j=1}^{J} \phi_{ij}} \tag{4}$$

where w_{ij} indicates whether cell (*i.j*) is covered at the end of mission *k* and Φ_{ij} indicates whether cell (*i.j*) belongs to the affected area. The total average coverage α over *N* missions is calculated as:

$$\alpha = \frac{1}{N} \times \sum_{k=1}^{N} \alpha_k \tag{5}$$

In this paper, the task of UAV swarm coverage is to maximize the coverage of the affected area and land safely. In order to comprehensively consider the coverage rate and the landing rate, this paper adopts the landing coverage rate (the coverage rate of the affected area under the premise of safe landing, collection ratio and landed (CRAL) as the evaluation index of the system. When *N* coverage missions are performed, the landing coverage rate of the *kth* mission is expressed by γ_k as follows:

$$\gamma_k = l_k \times \alpha_k = \begin{cases} \alpha_k, & l_k = 1 \\ 0, & l_k = 0 \end{cases} \tag{6}$$

The average landing coverage over *N* coverage missions is denoted by γ as:

$$\gamma = \frac{1}{N} \times \sum_{k=1}^{N} \gamma_k \tag{7}$$

Equations (5)–(7) define average metrics over *N* missions for evaluation (mean CR and mean CRAL). The optimization in Eq. (8) is stated per mission in expectation; maximizing the expected per-mission coverage (under safe landing constraints) is equivalent to maximizing its empirical average over many test missions. In this chapter, we aim to explore a strategy to optimize the maximum coverage rate of UAVs under the constraint of safe landing. The objective is to maximize the coverage of the disaster-stricken area while ensuring a safe landing within the constraints of the UAV's battery life. Assuming the coverage strategy of the UAV swarm is denoted as μ, the constrained optimization problem can be formulated as:

$$\begin{aligned} \max_{\mu} \quad & \frac{\sum_{i=1}^{I} \sum_{j=1}^{J} C_{ij} \phi_{ij}}{\sum_{i=1}^{I} \sum_{j=1}^{J} \phi_{ij}} \\ \text{s.t.} \quad & N = \sum_{k=1}^{N} l_k \\ & B_m^l \geq 0 \end{aligned} \tag{8}$$

4 Coverage path planning algorithm design

4.1 Deep reinforcement learning

This paper employs a value function-based Deep Double Q-Network (DDQN) for UAV swarm path planning. The method involves using environmental factors-like UAV swarm locations, affected areas, and no-fly zones-as state spaces on a map, with features extracted via deep neural networks. The DDQN algorithm evaluates actions in the current state, interacting with the environment in real time. States, actions, and rewards are recorded in replay memory. Random samples from this memory are used to train the

value network, refining the algorithm through continuous environment interaction. The optimal action, yielding the highest reward, is then chosen for UAV deployment.

4.2 Partially observable Markov decision process

To address the aforementioned challenges, the problem is formulated as a partially observable Markov decision process (POMDP) [28], defined by the tuple $(S, A, P, R, \Omega, O, \gamma)$. In this formulation, S denotes the state space, A is the action space, and $P : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to \mathbb{R}$ denotes the transition probability function. The reward function $R : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to \mathbb{R}$ maps each state–action–next-state tuple to a real-valued reward. The observation space is denoted by Ω and $O : S \to \Omega$ reprents the observation function. The discount factor $\gamma \in [0, 1]$ controls the relative importance of long-term and short-term rewards. The POMDP framework is adopted to capture the inherent partial observability in post-disaster environments, including unknown obstacle distributions and incomplete coverage information. Rather than explicitly maintaining and updating a belief state, an observation-based approximation is employed, wherein each UAV makes decisions based on a structured observation that integrates locally sensed information with coarse-grained global cues.

We describe the underlying (hidden) state space as S and define an observation space Ω to capture partial observability.

$$S = \underbrace{D^{I\times J\times 3}}_{\text{Environment Map}} \times \underbrace{\Phi^{I\times J}}_{\text{Disaster Area}} \times \underbrace{\mathcal{W}^{I\times J}}_{\text{Coverage Area}} \times \underbrace{P^{n_{max}\times 2}}_{\text{Position}} \times \underbrace{\mathcal{B}^{n_{max}}}_{\text{Battery Level}} \tag{9}$$

To support variable UAV counts while keeping the network input dimension fixed, we set a maximum swarm size n_{max} and pad $p(t)$, $b(t)$ to $R^{n_{max}\times 2}$ and $R^{n_{max}}$ with an availability mask. Note that the element $s(t) \in S$ can be denoted as

$$s(t) = (D, \Phi(t), \mathcal{W}(t), p(t), b(t)) \tag{10}$$

In this paper, $W(t)$ is a fused team-level coverage map available through the communication/map-fusion mechanism described in (centralized-information assumption).

In practice, the agent receives an observation $O(s(t)) \in \Omega$ rather than the full hidden state. We define

$$O(s(t)) = (D, \hat{\Phi}(t), \hat{W}(t), p(t), b(t)) \tag{11}$$

where D is a known static prior map, and $\hat{\Phi}(t)$, $\hat{W}(t)$ are belief/partial maps updated only within the UAV sensing footprint (unknown cells remain unobserved). The network takes $O(s(t))$ as input. Each UAV does not have access to a full-resolution global obstacle map. Instead, it is provided with a coarse-grained global map at reduced spatial resolution, while fine-grained obstacle information is acquired through onboard local sensing. Moreover, the disaster-affected regions are not fully known a priori and are progressively revealed as the UAVs explore the environment. Accordingly, at each time step, the observation is constructed as a multi-channel spatial representation that integrates the locally sensed obstacle map within the UAV's perception range, a downsampled global prior map and the current coverage status, and (iv) the discovered mask of the affected area.

- D: the environment map containing start and landing zones, no-fly zones, and obstacles, as shown in Fig. 2;
- $\Phi(t)$: the Disaster Area at time t;
- $W(t)$: the Coverage Area at time t, denoted as $W(w_1, w_2, \ldots, w_n)$, where w_i denotes the two-dimensional matrix for each UAV's coverage area.
- $p(t)$: the UAV's position at time t;
- $b(t)$: the UAV's remaining movement budget at time t.

The action space A is defined as a discrete set of motion and termination actions that control the UAV's movement in three-dimensional space. To avoid ambiguity, each action has a unique and well-defined physical meaning.

$$A = \begin{Bmatrix} North, South, East, West, \\ Ascend, descend, Landed \end{Bmatrix} \tag{12}$$

here $\{North, South, East, West\}$ denote horizontal moves to the 4-neighborhood grid cells in the x–y plane, corresponding to movement in the positive y, negative y, negative x, and positive x directions, respectively. $\{Ascend, Descend\}$ change the UAV altitude by one discrete level to enable obstacle overflight. Since heading/yaw is not modeled, 'forward/backward' are not separate actions and are represented by the grid-coordinate moves (North/South). The action Land is only valid when the UAV is in a designated landing cell and terminates the episode. Each action incurs a deterministic amount of energy consumption determined by the corresponding displacement. Specifically, a horizontal movement action results in an energy cost of b, whereas a vertical movement action incurs a higher energy cost of ξb.

The action set includes an explicit *Land* action, which terminates the episode only when the UAV is located at a designated start/landing cell. To avoid infeasible terminations, we apply an action-feasibility rule (action masking): *Land* is enabled only when the current cell belongs to the feasible landing set, and otherwise it is masked out. In addition, entering no-fly cells is penalized to discourage unsafe transitions. This clarifies that safe landing is not enforced purely by reward shaping, but also by termination feasibility.

In addition, the termination action *Land* is only feasible when the UAV is located at a designated landing cell, i.e., $(i, j) \in \mathcal{B}$. During training and evaluation, we apply an action-feasibility rule (action masking) to prevent invalid landing attempts outside $\mathcal{B}$.

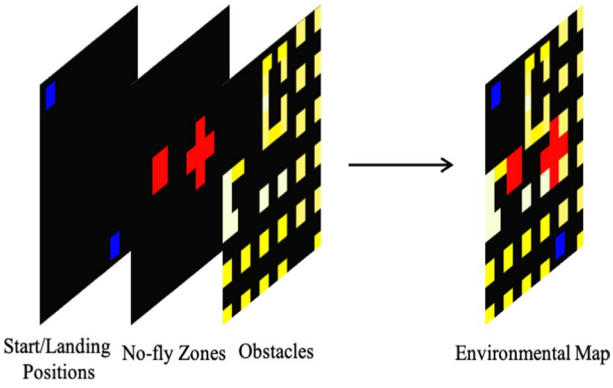


Fig. 2 Multichannel spatial composite map. Blue square denotes the start/landing position, red square denotes the no-fly zones and yellow square denotes the obstacles

This makes safe landing partially enforced at the action/termination level, while reward shaping (e.g., crash penalty) further accelerates learning toward safe behaviors.

The generalized reward function $R(s(t), a(t), s(t+1))$ consists of the following elements:

- r_c (reward for covering affected areas): At time *t*, a reward is granted for covering a new affected area, with η_c as a constant factor.

$$r_c = \eta_c \times \left(\sum \mathcal{W}_{t+1} - \sum \mathcal{W}_t \right) \tag{13}$$

- r_n (penalty for no-Fly zones): If the UAV's location $\omega_{ij} = 1$ falls within a no-fly zone, it incurs a substantial penalty, with η_n as a fixed constant.

$$r_n = \begin{cases} \eta_n, & \omega_{ij} = 1 \\ 0, & \omega_{ij} = 0 \end{cases} \tag{14}$$

- r_r (redundancy/overlap penalty): To discourage multiple UAVs from repeatedly covering the same cells, we penalize overlap among their coverage indicators. Let $w_i(t) \in \{0,1\}^{I \times J}$ denote the (binary) coverage indicator of UAV *i* at time *t*. We define the overlap penalty as

$$c(t)_{uv} = \sum_{i=1}^{n} w_i(t)_{uv} \tag{15}$$

$$r_r(t) = \eta_r \sum_{u=1}^{I} \sum_{v=1}^{J} max(0, c(t)_{uv} - 1) \tag{16}$$

where $w_i(t)_{uv} = 1$ means cell (*u*, *v*) is covered by UAV *i* (or newly covered, depending on the implementation). This term becomes positive whenever any two UAVs overlap, thereby encouraging implicit task partitioning.

- r_{mov} (movement penalty): A fixed movement penalty, represented by constant η_{mov}, curtails unnecessary reciprocal movements of the UAV.

$$r_{mov} = \eta_{mov} \tag{17}$$

- r_{crash} (crash penalty): The UAV must return and land safely in the landing zone before its battery runs out, or it incurs a significant penalty. This mechanism ensures optimal coverage of the affected area while prioritizing a safe landing.

$$r_{crash} = \begin{cases} 0, \mathcal{B}^l \geq 0 \wedge b_{ij} = 1 \wedge w_{ij} = 1 \wedge \psi_{ij} = 0 \\ \eta_{crash}, \quad \mathcal{B}^l < 0 \end{cases} \tag{18}$$

where (i,j) denotes the landing cell index.

4.3 Safety modeling: feasibility constraints versus reward shaping

We emphasize that safe landing in this work is evaluated through explicit feasibility checks at the mission level (battery feasibility and landing-cell feasibility), which determine whether an episode is counted as successfully landed. The reward design

complements this feasibility modeling: penalties such as r_{crash} discourage unsafe termination or battery depletion and help the value function learn safety-aware behaviors more efficiently. Therefore, safety is not enforced purely by reward shaping; instead, feasibility rules (e.g., valid landing conditions and termination feasibility) define mission success, while reward terms provide a learning signal that accelerates convergence to safe-and-efficient policies.

4.4 Model training

4.4.1 Map environment model

Figure 3a depicts the urban environment using a three-dimensional *X*-*Y*-*Z* Cartesian coordinate system, where the *X*-*Y* plane is the ground, and the *Z* axis represents height. Obstacle heights are mapped onto a planar terrain model using a grid method. This model is visualized in a heat map in Fig. 3b, showing buildings' heights with rectangles of varying color intensities. These colors correlate with the *X*-*Y* positions from 3(a). Each cell's color depth d_{ij} in the map is calculated as:

$$d_{ij} = \frac{h_{ij}}{\max h} \times \zeta \tag{19}$$

In Eq. (19), h_{ij} is the cell height, $\max h$ is the maximum height, and ζ is a scaling factor.

4.4.2 Neural network model

A convolutional neural network (CNN) is employed to extract features from multi-channel observation maps. As shown in Fig. 4, map data *D*, the affected area Φ, and the coverage area *W* are concatenated to form the multi-channel observation information $O(s(t))$. The CNN processes these features via global–local (downsampled) processing, before entering a fully connected layer that outputs action probabilities [29, 30]. To improve the algorithm's versatility and generalization, training involves varied environmental parameters like UAV count, battery levels, obstacle heights, and the dimensions and shapes of the affected areas. In this paper, we train with $n \in [2, 4]$ (City50 setting) and use the same network for any $n \leq n_{max}$; scaling beyond $nmax$ requires extending n_{max} and fine-tuning/retraining. Transfer learning accelerates training: the initial phase uses minor environmental alterations and ample battery for comprehensive UAV swarm

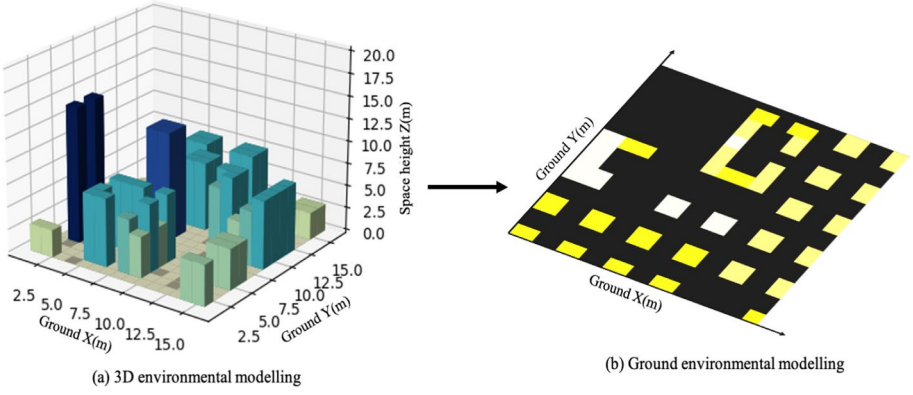


Fig. 3 Obstacle mapping. The depth of the yellow cell reflects the height of the obstacle, while black indicates no obstacle exists in that cell

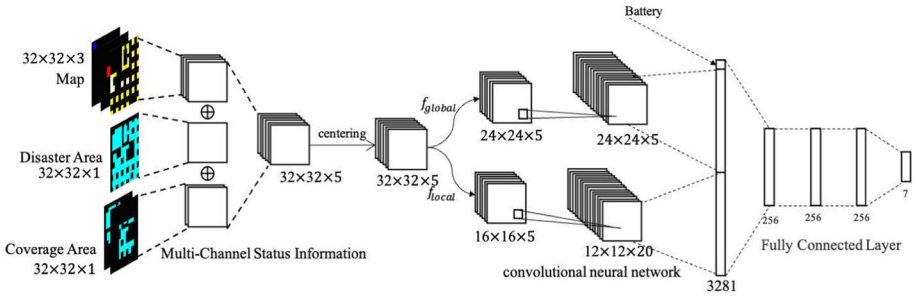


Fig. 4 Multichannel convolutional mapping network for integrated environmental analysis

coverage, familiarizing them with the environment. The model is then fine-tuned to handle larger problems and more complex environmental dynamics.

In our simulator, D is static during an episode. The affected-area mask Φ is sampled at the beginning of each mission and kept fixed during that mission (written as $\Phi(t)$ for generality). The coverage channel $W(t)$ is updated at every time step according to UAV sensing/coverage results. For collaboration, each UAV additionally maintains a per-UAV coverage matrix $w_i(t)$, which is used to compute the overlap (redundancy) penalty.

4.4.3 Cooperative strategy

It should be clarified that the proposed SRL framework adopts a centralized training and decentralized execution (CTDE) paradigm. During the training phase, global coverage information is available to enable stable learning of coordinated policies among UAVs. In contrast, during the execution phase, each UAV operates in a fully distributed manner and makes decisions solely based on locally observable states. Notably, the coverage matrix w_i does not require continuous real-time global synchronization. In practical post-disaster scenarios, only coarse-grained or delayed coverage information from neighboring UAVs is sufficient, which can be obtained via low-rate broadcast or intermittent communication when communication links are available.

In urban post-disaster emergency rescue missions, UAV path planning is not limited to finding the shortest flight trajectory but must also ensure extensive coverage of the disaster-affected area. This inherently leads to a multi-objective optimization problem, where both individual energy efficiency and global coverage performance must be jointly considered. As a result, a mixed dynamic of cooperation and competition naturally arises among UAVs. To promote cooperative behavior while mitigating excessive competition, this paper proposes a deep reinforcement learning algorithm with explicitly designed collaborative mechanisms.

Sequential execution mechanism: In the simulation environment, UAVs execute actions sequentially (i.e., $U_i \rightarrow U_{i+1}$). To avoid any informational advantage for later-acting agents, each UAV's observation at time step t is constructed from a snapshot of the environment at the beginning of that step (i.e., the observation does not include intermediate state updates caused by previously acting UAVs within the same step). Specifically, each UAV observes only its local sensing footprint and a coarse-grained global prior map, rather than the fully updated global coverage/state after earlier agents act. With parameter sharing across all agents, all UAVs are trained to follow the same policy under the same observation model; meanwhile, collaborative effects are enforced via environment-level delayed reward computation (e.g., overlap-related terms) based on

the team outcome. This design mitigates order-induced asymmetry and maintains fairness and stable policy convergence despite sequential execution.

Specifically, multichannel map representations are augmented with dynamic coverage information that records the regions explored by each UAV. After executing a flight action according to the state transition function, each UAV evaluates the overlap in newly covered areas and computes a collaborative reward based on (16). An illustrative example of overlapping coverage resulting from a UAV movement is depicted in Fig. 5.

The collaboration strategy is calculated as follows.

- At time T, given the state S_t of the environment, UAV U_i performs action a_t.
- During the coverage phase, each UAV U_i seeks to maximize its reward by covering adjacent areas, updating its coverage matrix accordingly, $w_i(t) = w_i(t+1)$.
- Individual rewards, based on the covered area, are calculated as: $R_i = r_c + r_n + r_{mov}$.
- Collaboration rewards are determined from the coverage matrix W, calculated as: $r_r = \eta_r \sum_{u=1}^{I} \sum_{v=1}^{J} max(0, c(t)_{uv} - 1)$.
- The collaboration strategy aims to maximize group rewards and promote cooperation by adjusting rewards: $R_i = R_i - r_r$.
- At time $T+1$, the environment state updates to $S_{(t+1)}$, and UAV U_{i+1} executes the next action $a_{(t+1)}$.

Although DDQN is a single-agent RL algorithm, our state includes the full swarm configuration (positions/battery of all UAVs) and the coverage matrices, so the learned Q-function is conditioned on the joint state. We execute actions sequentially among UAVs; hence each UAV's action changes the shared environment state for the next UAV, preserving inter-UAV coupling. Cooperation is induced by penalizing overlapping coverage through $R_i = R_i - r_r$, leading to division of labor. The specific pseudocode of Coverage path planning algorithm is shown as Algorithm 1. It describes decentralized execution with parameter sharing, where each UAV selects actions based solely on its local observation, while collaborative rewards are computed at the environment level.

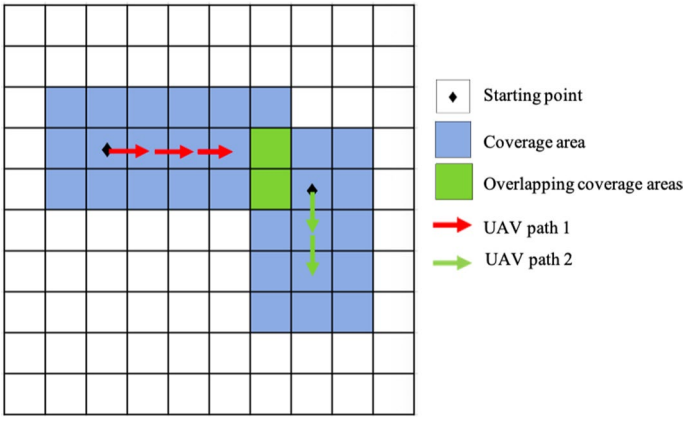


Fig. 5 UAV overlapping coverage area map

Input: Obstacle map D, affected-area mask Φ, initial UAV states $\{(x_0^u, y_0^u, z_0^u, B_0^u)\}$, action space $\mathcal{A}$, discount factor γ, learning rate α, replay buffer capacity $|\mathcal{D}|$, target network update interval T

Output: Trained DDQN parameters θ

online Q-network Q_θ and target network Q_{θ^-} with $\theta^- \leftarrow \theta$
Initialize replay buffer $\mathcal{D}$
Share network parameters θ among all UAVs
for Each training episode $\in 1 : Ep_{max}$ **do**
  Sample affected-area mask Φ
  Reset UAV states $(x_0^u, y_0^u, z_0^u, B_0^u)$ for $u = 1, 2, ..., n$
  Initialize coverage map $W(0)$ and per-UAV coverage maps $\{w_u(0)\}$
  for time step $t = 0, 1, ...$ **do**
    for each UAV $u = 1, ..., n$ **do**
      Construct observation $o_t^u = \text{Concat}(D, \Phi, W(t))$ in replay buffer $\mathcal{D}$
      Select action $a_t^u \in \mathcal{A}$ using ϵ-greedy policy based on $Q_\theta(o_t^u, a)$
      Execute action a_t^u, obtain next state and remaining energy B_{t+1}^u
    end for
    Update global coverage map $W(t+1)$ and per-UAV maps $\{w_u(t+1)\}$
    Compute individual reward r_t^u and collaborative reward r_t^r
    for each UAV $u = 1, ..., n$ **do**
      Store transition $(o_t^u, a_t^u, r_t^u + r_t^r, o_{t+1}^u)$ in $\mathcal{D}$
    end for
    Sample mini-batch from $\mathcal{D}$
    Update θ by minimizing the DDQN loss:
      $$L(\theta) = \left(r + \gamma Q_{\theta^-}(o', \arg\max_{a'} Q_\theta(o', a')) - Q_\theta(o, a) \right)^2$$
    if $t \bmod T = 0$ **then**
      Update target network: $\theta^- \leftarrow \theta$
    end if
    if all UAVs have executed the Land action or $B_t^u = 0$ **then**
      Terminate episode
    end if
  end for
end for
return

Algorithm 1 SRL-based coverage path planning algorithm for UAV Swarm

4.5 Rationale for choosing DDQN

The selection of DDQN is motivated by both the characteristics of the problem and practical implementation considerations. The proposed framework adopts a discrete action space with constrained motion primitives and explicit termination actions, which naturally favors value-based reinforcement learning methods. Compared with standard DQN, DDQN effectively alleviates overestimation bias, resulting in more stable learning behavior in safety-critical settings. Although policy-gradient methods such as PPO and SAC are well suited for continuous action spaces, they typically require careful tuning of entropy and clipping parameters.

We acknowledge that DDQN is originally a single-agent value-based algorithm, while our task involves multiple UAVs. In this work, DDQN is employed in a parameter-sharing manner: all UAVs share the same Q-network and are trained on the union of transitions collected from multiple agents interacting with a shared environment. This corresponds to independent learners with shared parameters and can be viewed as CTDE-style training with decentralized execution, where each UAV acts based on its

own partial observation, and training aggregates multi-agent experiences to improve a common policy.

We adopt DDQN mainly because the action space in our coverage-and-landing task is discrete (including an explicit *Land* termination action), and (DDQN offers stable learning and good sample efficiency in discrete domains. Nevertheless, we emphasize that this choice comes with limitations compared with dedicated MARL frameworks. In particular, parameter-shared DDQN does not explicitly model a centralized critic or value decomposition (as in QMIX/VDN), and may be less expressive for learning complex coordination strategies than actor-critic MARL approaches (e.g., MADDPG or PPO-based methods). Addressing these limitations by integrating value-decomposition or centralized-critic MARL frameworks is an important direction for future work.

5 Simulations

5.1 Simulation setup

This study implements a deep reinforcement learning–based simulation environment using Python 3.9 and TensorFlow 2.5.0, and constructs two urban scenarios with different spatial scales, namely *City32* and *City50*. The UAVs operate at a fixed flight altitude $h^{'}$, which enables ground perception over local grids of size 5×5 or 3×3. The key network and training parameters are summarized in Table 1. To ensure a fair comparison, all algorithms are evaluated under identical environmental conditions, including the same obstacle layouts, disaster-affected regions, UAV initial positions, and random seeds. The experimental evaluation is designed to demonstrate the feasibility and practical effectiveness of the proposed framework under representative urban post-disaster scenarios.

5.2 Baseline discussion

The selected baselines are chosen to provide representative comparisons across different levels of decision-making complexity rather than to exhaustively benchmark all state-of-the-art methods. Specifically, BSM serves as a heuristic baseline without learning capability, highlighting the advantages of learning-based decision-making; K-DQN represents a DQN-based approach with handcrafted coordination rules; and GLRL reflects a global-reward-driven learning strategy without explicit collaborative modeling. We acknowledge that these baselines are not state-of-the-art. However, direct comparison with many recent methods is challenging due to substantial differences in problem

Table 1 Hyperparameters

Parameter	*City32*	*City50*	Description
l	17	17	Local map size
g	3	5	Global map downsizing
n_c	3	3	Number of conv. layers
n_k	16	16	Number of kernels
s_k	5	5	Conv. kernel width
F	256	256	Flatten layers size
η_c	3	1	Reward coefficient for covering affected areas
η_n	-8	-5	Penalty coefficient for no-Fly zones
η_r	-0.7	-0.2	Overlap penalty coefficient
η_{mov}	-2	-1	Movement penalty coefficient
η_{clash}	-12	-10	Crash penalty coefficient

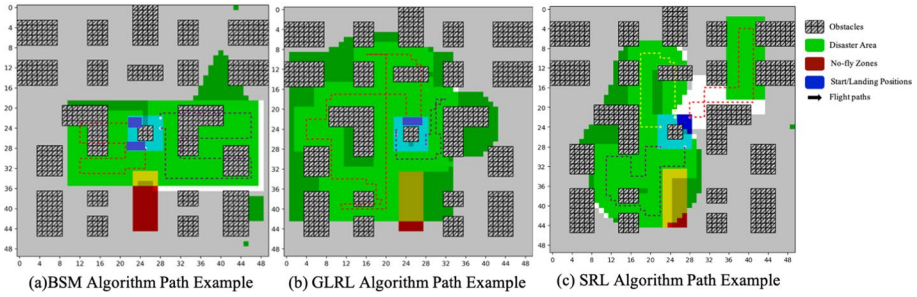


Fig. 6 Only shows the differences in the path-planning form in *City50* scenario

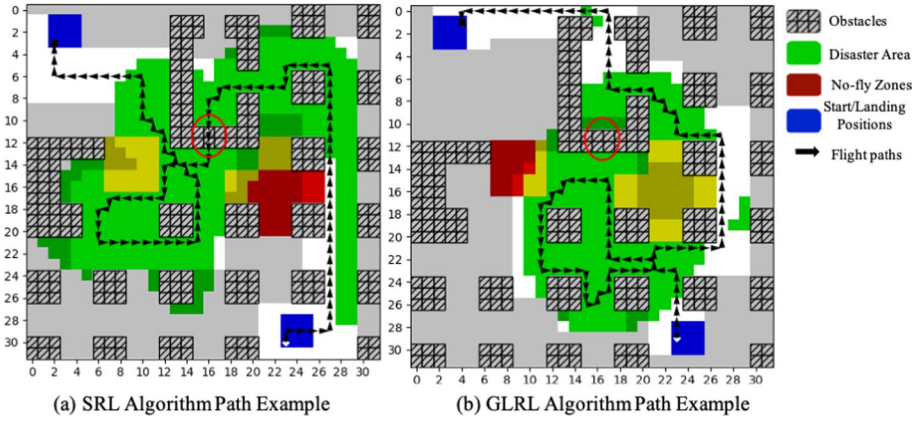


Fig. 7 Paths generated by SRL and GLRL in *City32* scenario

formulation, action spaces, and safety constraints. In particular, most state-of-the-art approaches focus on continuous low-level control or collision avoidance, whereas our work addresses mission-level safe landing and coverage planning with discrete actions and explicit termination conditions.

5.3 General evaluation

In this subsection, we evaluate the trajectory planning performance of the proposed algorithm under different scenarios (City32 and City50). For clarity and consistency in visualization across scenarios, the following legend conventions are adopted throughout this section: red arrows indicate the trajectories generated by the comparison algorithm, while black arrows represent the trajectories of the proposed algorithm. For clarity and consistency in visualization across scenarios (Figs. 6, 7), green squares denote disaster-affected areas, red squares denote no-fly zones, and yellow squares denote the overlap between disaster-affected areas and no-fly zones; blue squares indicate landing locations and gray squares indicate obstacles.

5.3.1 City50

In the *City50* scenario, the disaster-affected area comprises 20–50% of the total area, with 2–4 UAVs operating at battery levels of 100–150.

This paper introduces a Collaborative Strategy Reinforcement Learning algorithm (SRL) based on the DDQN algorithm, with an added collaborative strategy. To verify

Table 2 Performance metrics averaged over 1000 random scenario Monte Carlo iterations

Metric	SRL	GLRL	K-DQN	BSM
Landed	0.992	0.981	0.956	0.612
CR	0.856	0.815	0.782	0.572
CRAL	0.842	0.801	0.753	0.38

Table 3 Performance metrics averaged over 1000 random scenario Monte Carlo iterations

Metric	SRL	GLRL	K-DQN
Landed	0.991	0.985	0.965
CR	0.756	0.710	0.694
CRAL	0.743	0.703	0.683

the effectiveness of this algorithm, the K-means algorithm and DQN model combination method (K-DQN), GLRL, and BSM were used as comparison algorithms.

The K-DQN method refers to using the K-means algorithm to divide the target area into different clusters in the task of multi-UAV coverage path planning, and then utilizing the DQN algorithm to achieve single-UAV path planning within each cluster.

Table 2 compares the safe landing rate, coverage rate, and landing coverage of UAVs using BSM, GLRL, and SRL. The SRL algorithm, as proposed in this paper, surpasses the others in landing and coverage rates.

Block scan method (BSM): This algorithm divides the target area into left and right sections, with each of two UAVs covering a different section. Figure 6a shows the BSM's coverage path.

Global local deep reinforcement learning (GLRL): GLRL enables UAVs to quickly adapt to the environment and plan paths using both global and local information. However, it lacks altitude adjustment, leading UAVs to circumnavigate obstacles and consume more battery. Figure 6b shows the GLRL coverage path.

Synergy deep reinforcement learning (SRL): SRL enhances GLRL with a collaborative strategy, allowing better coordination and labor division among UAVs. It also includes altitude adjustment, enabling UAVs to fly over obstacles. Figure 6c shows the SRL coverage path.

5.3.2 City32

In the *City32* scenario, 20–50% of the area is disaster-affected, with UAV battery level set at 50–150. This scenario, which is denser in tall buildings than *City50*, favors flying over rather than around obstacles. Figure 7a shows the SRL algorithm's path, efficiently covering the lower-left area by flying over barriers, as highlighted in a red circle. This method reduces flight path length compared to circumventing obstacles. Figure 7b illustrates the GLRL algorithm's coverage path. Table 3 compares the safe landing rate, coverage rate, and landing coverage rate of UAVs for GLRL, K-DQN , and SRL algorithms. The SRL algorithm, detailed in this paper, demonstrates higher performance in both landing and coverage rates compared to others.

5.4 Behavior analysis of learned UAV policies

Beyond quantitative performance metrics, this section provides qualitative insights into the behaviors learned by the proposed SRL framework and analyzes how environmental factors and reward design jointly influence UAV trajectories and landing decisions.

- *Impact of obstacle density:* Obstacle density plays a dominant role in shaping UAV motion patterns. In relatively sparse environments (e.g., City32), UAVs tend to adopt more direct trajectories toward uncovered regions, prioritizing horizontal exploration to maximize coverage efficiency. In contrast, in denser urban environments (e.g., City50), UAVs exhibit smoother and more conservative trajectories, frequently adjusting altitude to bypass congested obstacle clusters. This behavior reflects a learned trade-off between path length and collision risk, where the policy implicitly favors safer altitude transitions over aggressive low-altitude navigation in narrow corridors.
- *Influence of battery constraints:* Battery limitations significantly affect both exploration range and landing decisions. When the remaining energy is sufficient, UAVs demonstrate proactive exploration behavior, extending their trajectories toward distant uncovered regions. As battery levels decrease, however, the policy increasingly biases actions toward nearby regions and feasible landing zones. This transition is particularly evident near the end of each episode, where UAVs avoid risky detours and prioritize energy-feasible return paths. These behaviors indicate that the energy-aware reward and termination conditions effectively guide the policy to balance coverage objectives with safe landing requirements.
- *Role of collaborative reward design:* The collaborative reward mechanism strongly influences inter-UAV coordination. By incorporating coverage overlap information, the learned policy encourages spatial dispersion among UAVs, reducing redundant exploration. As a result, UAVs dynamically adjust their trajectories to avoid regions already covered by neighboring agents, even without explicit communication of planned paths. This emergent coordination is more pronounced in complex environments, where decentralized yet cooperative behavior leads to improved coverage ratio at landing (CRAL) despite increased environmental uncertainty.
- *Interaction between factors:* Importantly, the observed behaviors do not arise from a single factor in isolation, but from the interaction between obstacle constraints, energy limitations, and collaborative incentives. For instance, in obstacle-dense regions with low remaining battery, UAVs tend to favor conservative altitude adjustments and early landing decisions, while collaborative rewards prevent multiple UAVs from converging on the same landing areas. This synergy explains why the proposed SRL framework maintains robust performance across different city layouts, whereas baseline methods often suffer from either excessive collisions or inefficient coverage.

Overall, these observations suggest that the proposed framework does not merely optimize numerical metrics, but learns structured and interpretable behaviors that align with the practical requirements of post-disaster UAV operations, including safe navigation, energy-aware planning, and decentralized cooperation.

5.5 Hyperparameter sensitivity and robustness analysis

The learned policy exhibits adaptive behavior across varying system and environmental conditions.

Increasing the number of UAVs intensifies coordination complexity and potential coverage redundancy, which is effectively mitigated through collaborative rewards that

promote spatial dispersion; with fewer UAVs, the policy becomes more conservative, favoring localized exploration and earlier landing due to limited collective sensing.

Battery capacity further governs the trade-off between exploration and safety, where larger energy budgets enable broader coverage via longer trajectories, while constrained batteries induce risk-averse behavior and earlier mission termination, facilitated by energy-aware reward penalties.

Obstacle density also plays a critical role: in sparse environments, UAVs follow more direct paths, whereas denser obstacles encourage smoother trajectories with altitude adjustments to maintain collision avoidance, without compromising safe landing performance. Although static obstacles are assumed per episode, the framework inherently operates under partial observability using local sensing and coarse global information, allowing straightforward extension to dynamic or uncertain environments through real-time observation updates.

Finally, while direct comparison with frameworks such as Dawn [19] is nontrivial due to differences in action space and safety modeling, the proposed SRL framework complements such approaches by addressing mission-level coverage and safe landing with discrete decisions and termination conditions, and can be naturally integrated with lower-level controllers for practical deployment.

6 Practical deployment considerations and trade-off analysis

Although the proposed SRL framework is evaluated in simulation, it is important to discuss the challenges and considerations that may arise during real-world UAV deployment in post-disaster environments.

6.1 Impact of sensor noise and partial observability

In practical deployments, onboard sensors such as LiDAR, cameras, or depth sensors inevitably suffer from noise, occlusion, and limited sensing range. These factors may lead to imperfect obstacle detection and delayed perception of affected areas. In the proposed framework, such uncertainties are partially mitigated by the observation-based decision-making paradigm, where UAVs rely on local sensing combined with coarse global information rather than full state knowledge. Nevertheless, severe sensor noise may degrade the accuracy of the local maps, potentially resulting in conservative trajectories or delayed landing decisions. Incorporating probabilistic occupancy maps or belief-state filtering could further improve robustness in real deployments.

6.2 Communication latency and decentralized coordination

Post-disaster environments are often characterized by disrupted or unreliable communication infrastructure. The proposed framework adopts decentralized execution with parameter sharing, which reduces reliance on continuous inter-UAV communication during operation. However, communication latency or packet loss may still affect the timeliness of shared coverage information, leading to temporary redundancy or suboptimal coordination. In practice, this issue can be alleviated by asynchronous updates, delayed reward aggregation, or periodic synchronization strategies, which are compatible with the current learning framework.

6.3 Trade-off between safe landing and coverage efficiency

A key practical challenge lies in balancing safe landing constraints with coverage efficiency. Enforcing conservative safety constraints, such as early landing thresholds or strict energy margins, improves landing reliability but may reduce coverage performance. Conversely, aggressive exploration strategies improve coverage at the risk of energy depletion or unsafe termination. In the proposed framework, this trade-off is implicitly regulated through reward weighting and termination conditions. By adjusting the relative importance of energy-related penalties and coverage rewards, the policy learns to adapt its behavior according to mission priorities. This mechanism provides practical guidance for deployment, allowing system designers to tune the balance between safety and efficiency based on operational requirements.

Overall, while real-world deployment introduces additional uncertainties and constraints, the proposed framework offers a flexible decision-level solution that can be extended with more advanced perception, communication, and control modules, paving the way toward practical post-disaster UAV applications.

7 Conclusions

In this paper, a novel deep reinforcement learning approach has been proposed for multi-UAV coverage path planning in complex urban environments, with particular emphasis on post-disaster emergency rescue scenarios. The proposed approach utilizes a convolutional neural network to extract environmental data, integrating this with a coverage matrix to compute collaborative rewards during reinforcement learning training. This method effectively guides real-time decision-making within UAV swarms, emphasizing the optimization of the shortest coverage paths and rewards maximization. The paper demonstrates how UAV swarms can achieve maximal coverage in urban post-disaster rescue operations, showcasing both generality and strong generalization capabilities. Through simulation experiments, the performance of three algorithms is comparatively analyzed in the context of a rescue mission. The main theoretical contribution of this work lies in the unified modeling of coverage efficiency and landing safety. Instead of optimizing coverage and considering safety as an auxiliary constraint, the proposed framework integrates safe landing into the reward structure and performance metrics. This ensures that path planning decisions inherently balance exploration, energy consumption, and return feasibility, which is particularly critical in post-disaster urban environments.

Author contributions

Lei Gao: Conceptualization, Methodology, Validation, Software, Writing-original draft. Xin Xiang: Methodology, Validation, Writing—review and editing. Haiyan Zhao: Writing—review and editing, Resources. Huie Chen: Writing—review and editing. Yuanai Xie: Methodology, Resources, Writing—review and editing. All authors have read and agreed to the published version of the manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication
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